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# Brawn before bite in endemic Asian mammals after the end-Cretaceous extinction

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## eLife Assessment

This **important** study fills a major geographic and temporal gap in understanding Paleocene mammal evolution in Asia and proposes an intriguing "brawn before bite" hypothesis grounded in diverse analytical approaches. However, the findings are **incomplete** because limitations in sampling design - such as the use of worn or damaged teeth, the pooling of different tooth positions, and the lack of independence among teeth from the same individuals - introduce uncertainties that weaken support for the reported disparity patterns. The taxonomic focus on predominantly herbivorous clades also narrows the ecological scope of the results. Clarifying methodological choices, expanding the ecological context, and tempering evolutionary interpretations would substantially strengthen the study.

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## Abstract

The first 10 million years (Myr) following the Cretaceous-Paleogene (K-Pg) mass extinction marked a period of global greenhouse conditions and dramatic rise of placental mammals. Because ~80% of known terrestrial sections capturing post-K-Pg mammal recovery come from North America, a substantial knowledge gap exists in the tempo and mode of recovery in Asia, where only 3% of sites are located and most contain species found nowhere else. We show that isolated Paleocene placental assemblages from China (1) reached high tooth size disparity early in the Paleocene, (2) tracked regional and global environmental changes in their dental shape later in the Paleocene, and (3) achieved maximum dental shape-performance integration near the end of the first 10 Myr post-K-Pg. This 'brawn before bite' transformation, coupled with prolonged dental shape versus performance variability, favors a scenario whereby many living orders of placental mammals were borne out of phenotypically and functionally plastic ancestral assemblages, including those in tropical south China, during the Paleocene.

## Main text

The Cretaceous-Paleogene (K-Pg) mass extinctions accelerated the formation of modern-day global biota. In particular, the more than 6,000 species of living placental mammals trace their origins to the diversification of major orders around or after the K-Pg boundary <sup>1-3</sup>. Different hypotheses (for example: long-fuse, explosive, and short-fuse) about the timing of their adaptive radiation have been variably supported by analyses of molecules, fossils, and model-based approaches <sup>1,4-6</sup>. Regardless of the diversification scenarios favored by the competing explanations, the Paleocene epoch (66-56 Myr ago) has been highlighted as the most critical time interval for establishing phylogenetic and rate parameters for the placental adaptive radiation <sup>5,7</sup>. However, substantial

asymmetry exists in the quantity and quality of terrestrial fossil records that document the first 10 Myr of the ‘Age of Mammals’, with ~80% of known terrestrial K-Pg boundary sections occurring in North America <sup>8</sup>. Earliest Paleocene mammals are not known from Europe <sup>9,10</sup>, rendering detailed reconstructions of post-K-Pg recovery in that continent difficult. The post-K-Pg fossil assemblages from Asia have hitherto not been considered in analyses of post-K-Pg recovery dynamics <sup>11</sup>.

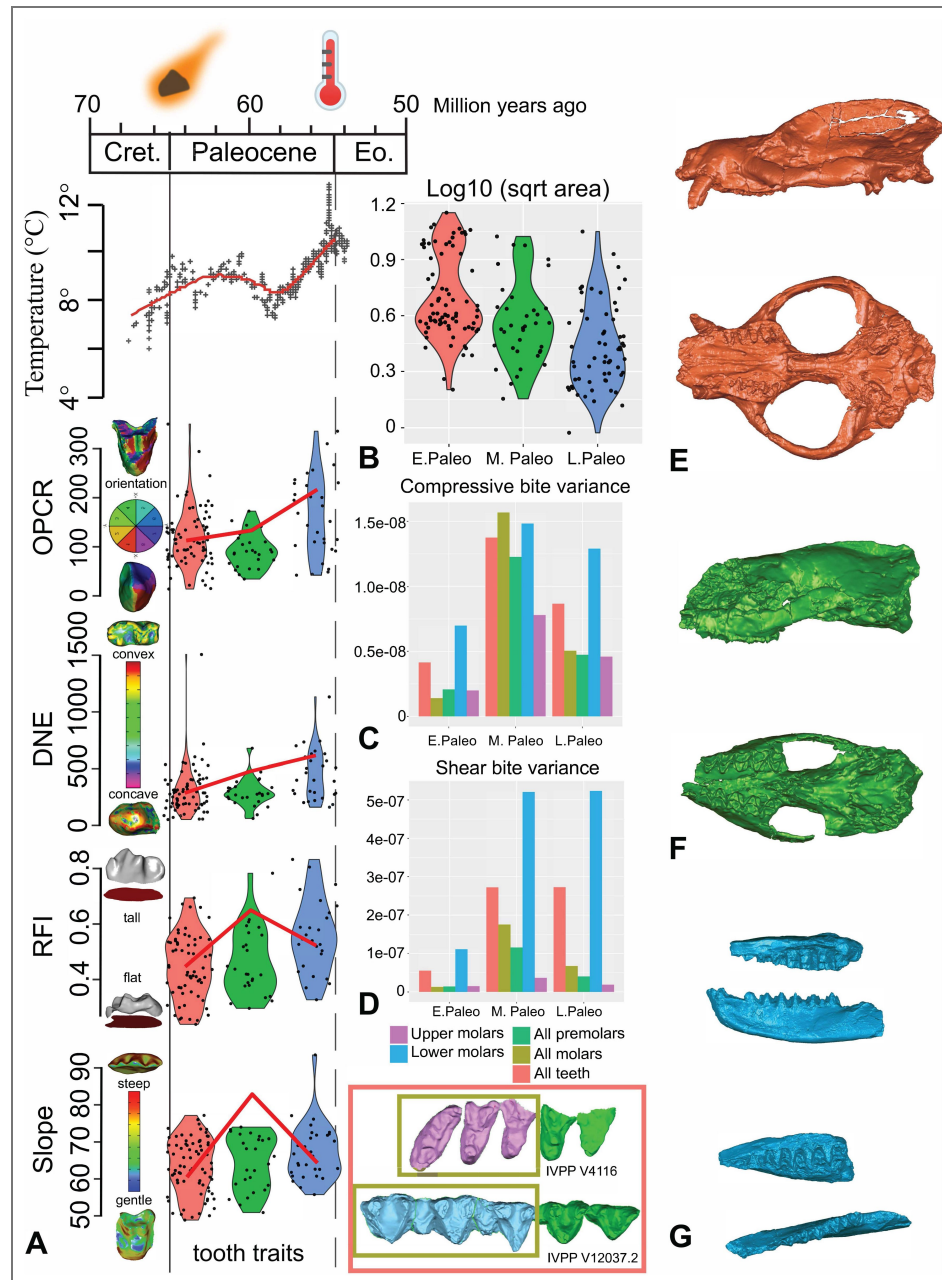
The North American fossil record suggests that post-K-Pg mammal taxonomic diversity recovery was relatively rapid at higher latitudes; most occurred within the first ~0.3 to 1 Myr of the Paleocene epoch <sup>12–17</sup>. This initial placental mammal diversification was driven by archaic groups (i.e., stem lineages and those with no living relatives), followed by the first appearance of many modern orders during two peak hyperthermal events in the past 66 Myr, at the end-Paleocene and early Eocene climatic maxima, respectively <sup>18,19</sup>. Distinct from these taxonomic recovery patterns, high selectivity of mammalian ecomorphological extinction across the K-Pg boundary indicates a primary productivity filter at the start of the Cenozoic Era (66 Myr ago to the present), disproportionately impacting large herbivores and their predators <sup>20</sup>. Additionally, the first 10 Myr of mammalian brain evolution after the K-Pg was marked by niche partitioning along a size gradient; endocranial traits reflecting more complex sensory processing did not appear until the Eocene (56 Myr ago). This phenomenon has been termed the ‘brawn before brains’ hypothesis <sup>21</sup>. A similar pattern in initial size-driven diversification followed by expansion of ecomorphological disparity is also observed in mammal jaws, indicating a general dynamic across phenotypic systems <sup>22</sup>. Whether this mode of evolution, whereby size disparity increase precedes other ecometric traits, should be understood as a global phenomenon during the post-K-Pg placental radiation remains untested, in large part because no such analyses have centered on non-North American continental fossil mammal records.

We investigated dental topography-performance shifts and timing of ecomorphological diversification by developing and leveraging the largest dataset to date of Paleocene Asian mammal assemblages. Our analyses focused on three of the most fossiliferous and biogeographically isolated Paleocene sedimentary sequences in paleotropical Asia: The Nanxiong, Qianshan, and Chijiang Basins in present-day south China <sup>23–27</sup> (Fig. S1 [↗](#)). We generated a new phenotypic dataset of 200 Asian Paleocene mammal teeth using high-resolution microcomputed tomography and laser scanning, capturing 37 species endemic to low-latitude east Asia and which are brought to bear on K-Pg recovery dynamics for the first time (Data S1–S2 [↗](#)). We used dentitions as ecological indicators <sup>28</sup> and examined temporal shifts in tooth crown complexity, curvature, and height and their association with tooth deformation resistance using topographic and simulation analyses. Our dataset spans the Paleocene, the first 10 Myr of the Cenozoic, enabling us to test the hypothesis that dental topography and tooth puncturing and shearing performance linkages showed delayed niche expansion relative to body size disparity increase during this initial period of post-K-Pg mammal recovery in Asia.

## Dental traits paralleled Paleocene global and regional environmental conditions

We found that dental topographic trait variability in Paleocene mammals in south China tracked global and regional climatic changes despite stasis in high-level taxonomic composition over the course of the first 10 Myr after the K-Pg transition (Fig. 1 [↗](#)). Dental height and sharpness variability were low in the beginning and end of the time interval, with a spike in the middle Paleocene corresponding to a short-lived negative excursion in global temperature. In contrast, elevated low-level taxonomic turnover of genera and species within the Paleocene indicates cladogenetic shifts, rather than anagenetic adaptation, underpin this dental topographic evolution (Data S1 [↗](#))<sup>11</sup>. These findings suggest that in addition to its impact on crown group mammals, the K-Pg extinctions and subsequent climatic fluctuations also filtered out archaic placental mammals with lower speciation rates in favor of rapidly speciating taxa <sup>4</sup>.

By contrast, there is no support for significant shifts in dental topographic trait mean values from the early to the middle Paleocene for the majority of analytical iterations of clade and dental data partitions (Figs 1A [↗](#), S4; Table 1 [↗](#); Data S7 [↗](#)). However, in most analyses we observed a



**Fig. 1. Temperature and fossil mammal dental trait shifts during the first 10 Myr of the Cenozoic.**

(A) Dental topographic trait values (boxplots) and mean variance (red curves) during the first 10 Myr of the Cenozoic, signifying the time after the K-Pg mass extinctions and before the Paleocene-Eocene hyperthermal event. Global temperature curve based on Zachos et al. <sup>58</sup>. Dental traits measured include crown complexity (OPCR, orientation patch count rotated), curvature (DNE, Dirichlet normal energy), height (RFI, relief index), and slope. (B) Mammal tooth size distributions represented by log 10 square root tooth area, in units of log10 millimeters. (C) Variance of compressive bite performance based on tooth crown finite element simulations, in units of squared Joules. (D) Variance of shear bite performance based on tooth crown finite element simulations, in units of squared Joules. Examples of endemic Asian fossil specimens analyzed: (E) Lateral and ventral views of early Paleocene Chinese endemic pantodont (CEP) *Bemalambda nanhsiungensis* IVPP (Institute of Vertebrate Paleontology and Paleoanthropology, Chinese Academy of Sciences) V4116. (F) Lateral and ventral views of middle Paleocene CEP *Harpyodus decorus* IVPP 5035.1. (G) Lateral and occlusal views of late Paleocene CEP *Guichilambda zhajii* IVPP V12037.2 (dentary) and V12037.3 (maxillary fragment). Firey asteroid symbols indicate the end-Cretaceous asteroid impact in the Yucatán Peninsula; thermometer symbols indicate the Paleocene-Eocene hyperthermal event. Subdivisions of the Paleocene approximately correspond to the Shanghuan, Nongshanian, and Gashatan Asian Land Mammal Ages, respectively (see supplemental text for competing age boundary scenarios).

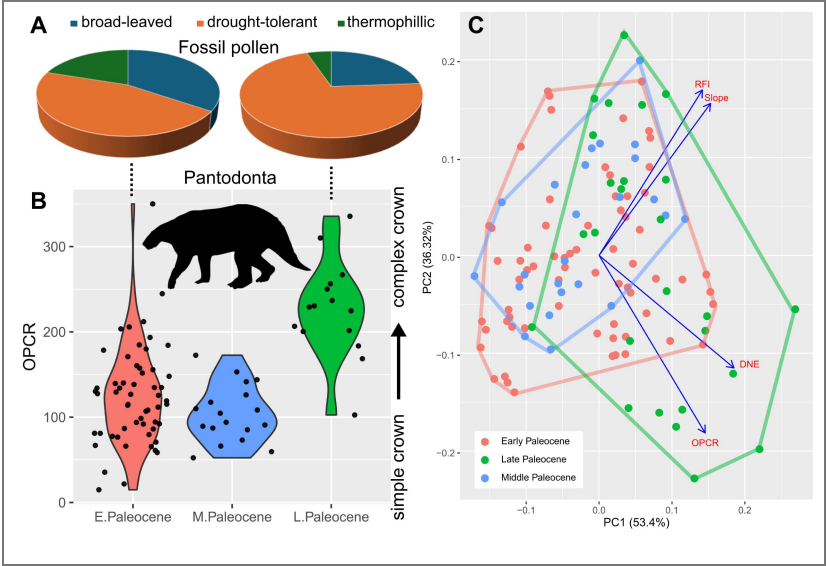
significant shift in at least one dental topographic metric from the middle to the late Paleocene (Table 1 [↗](#); Data S7 [↗](#)). The larger-bodied Chinese endemic pantodont (an ungulate-like archaic placental group; “CEP”) mammals tend to increase dental complexity (OPCR, orientation patch count rotated) and curvature (DNE, Dirichlet normal energy) whereas smaller, non-pantodonts (Chinese arctostylopids and anagalids; rodent-like archaic placental groups) in the dataset exhibited no significant dental traits shifts from the middle to late Paleocene (Fig. S4 [↗](#); Data S7 [↗](#)). The transformation by CEPs in complexity and curvature indices relates to the capacity of teeth to resist wear and coincides with temperature and aridity increase towards the end-Paleocene thermal maximum event (Figs. 1 [↗](#)–2 [↗](#)). Multiple geological and geochemical proxies suggest that paleoclimate in and around the Nanxiong Basin K-Pg section in south China reflects a latitudinally much broader global tropical zone during the Paleocene <sup>29</sup> relative to present-day Earth, as well as rapid shifts between more versus less humid intervals during the first 10 Myr of the Cenozoic <sup>30</sup>. Results from feldspar-quartz (F:Q) ratios, clay mineral composition analysis, diffuse reflectance spectroscopy (DRS), stable carbon isotope analysis, and total organic carbon (TOC) analysis all support this regional paleoclimate profile <sup>30,31</sup>. Local climate reconstructions for the latest Cretaceous indicate relatively warm and dry intervals, followed by warm and humid climates during the earlier Paleocene, then a return to less humid but still warm conditions in the later Paleocene <sup>31</sup>. Chinese endemic pantodont dentitions tracked shifts in this paleoenvironmental progression.

The overall increase in dental complexity and curvature also coincided with an increase in drought-tolerant flora in south China (Fig. 2 [↗](#)) and specifically with paleoenvironmental reconstructions in the Nanxiong Basin <sup>32</sup>. Palynological evidence suggests that in addition to a predominance of broad-leaved, deciduous plants mixed with a smaller percentage of conifers than in fossil localities further north <sup>33</sup>, south China also recorded drought resistant taxa such as the maidenhair fern *Pterisporites* in the early Paleocene Shanghu formation <sup>34</sup> and again in late Paleocene samples <sup>35</sup>. Expanded comparisons across time and space suggest fossil pollen taxa that are indicators of seasonal aridity increased from 20.3% of late Paleocene pollen samples to 34.3% of early Eocene pollen samples <sup>35</sup>. These shifts appeared to have stabilized by the end of the Eocene, when taxonomically modern floras became established in Asia <sup>36</sup>. In addition to tracking Paleocene temperature trends (Fig. 1 [↗](#)), Asian mammal dental topography (OPCR and DNE in particular) also mimicked trends in the marine realm, where global planktonic foraminifera records demonstrate a similar pattern in species richness curve, as well as with global  $\delta^{13}\text{C}$  <sup>37,38</sup> and south Pacific  $\text{CO}_2$  levels <sup>39</sup>. These broader associations offer strong indications that endemic Paleocene mammals in south China composed a dynamic assemblage evolving in step with regional and global environmental changes during the first 10 Myr after the K-Pg mass extinctions.

We detected a shift and expansion of mammal dental morphospace in the late Paleocene (Fig. 2 [↗](#)). An overall shift towards increased dental topographic trait magnitudes in late Paleocene samples is driven mainly by CEPs (Table 1 [↗](#)), despite the fact that they constitute a minority of the overall data (41% of teeth) as well as late Paleocene partition (21% of teeth). Additionally, dental metric disparity is significantly higher in the late Paleocene partition than in the two preceding time bins (Table 1 [↗](#)). This pattern is driven both by increased dental curvature and complexity (DNE & OPCR) in larger-bodied CEPs and crown height (RFI) in smaller-bodied non-pantodonts, in addition to differences in disparity among the clades (Data S7 [↗](#); bootstrapped variance for DNE, OPCR, and RFI are at least twice as large in non-pantodonts compared to CEPs). This suggests that an expansion of dental disparity in the late Paleocene occurred across the size spectrum of endemic Asian mammals. Over the same time interval examined, body size disparity appeared to have reached maximum saturation early on, in the early Paleocene (Fig. 1B [↗](#), S6A, Table 1 [↗](#), Data S5 [↗](#)), indicating that substantial increases in the disparity of dental complexity, curvature, and height lagged behind maximum size disparity during the Paleocene. This marked dental evolutionary pattern of ‘brawn before bite’ in endemic Asian mammals mirrors the endocranial and jaw functional morphology patterns identified in their North American and European counterparts <sup>21,22</sup>, underscoring that an initial size-driven post-K-Pg recovery followed by

**Fig. 2. Association of paleopalynological data from the Nanxiong Basin, south China, and late Paleocene niche expansion in endemic Asian fossil mammals.**

(A) Proportion of environmental humidity indicator taxa from early versus late Paleocene paleobotanical localities, respectively, in the Nanxiong Basin; data based on 34,35 (Data S12). (B) Boxplots of dental complexity (OPCR, orientation patch count rotated) in the Chinese endemic pantodont (CEP) data partition across the three Paleocene time intervals examined. Note the concomitant increase in CEP tooth complexity (OPCR) and increased proportion of drought-tolerant plant species in the Nanxiong Basin during the late Paleocene. (C) Principal component morphospace of all tooth data analyzed; convex hulls delineate overall morphospace occupation during each time interval. Eigenvectors of the four dental topographic traits are indicated in blue. Late Paleocene shift and expansion in dental topographic morphospace is statistically significant at the  $p = 0.05$  level (Table 1). Pantodont silhouette by S. Traver from [phylopic.org](https://www.phylopic.org).



**Table 1. Pairwise t test of dental topographic trait and disparity differences across adjacent time bins.**

Dental topographic trait differences are assessed across time intervals in all-data, Chinese endemic pantodont, and non-pantodont partitions. Dental trait disparity was estimated based on all principal component axes using the outputs of PCA. Tooth size variance differences were calculated from tooth area or square root of tooth area in all-data and no-outlier partitions to assess effect of outliers on statistical significance (see Data S6 for details). Bolded font indicates  $p$  values  $< 0.05$ .

| All data                           | Early-Middle Paleocene | Middle-Late Paleocene |
|------------------------------------|------------------------|-----------------------|
| DNE                                | 0.7753                 | <b>0.003</b>          |
| OPCR                               | 0.0902                 | <b>0.0001</b>         |
| RFI                                | 0.605                  | <b>0.0128</b>         |
| Slope                              | 0.643                  | 0.287                 |
| PCA Disparity                      | 0.4562                 | <b>0.0001</b>         |
| Tooth size                         | <b>&lt;0.0001</b>      | <b>&lt;0.0001</b>     |
| Chinese Endemic Pantodonts         |                        |                       |
| DNE                                | 0.9264                 | <b>0.0002</b>         |
| OPCR                               | 0.17                   | <b>&lt;0.0001</b>     |
| RFI                                | 1                      | 1                     |
| Slope                              | 1                      | 1                     |
| Chinese Arctostyloids and Anagalis |                        |                       |
| DNE                                | 0.847                  | 0.135                 |
| OPCR                               | 0.47                   | 0.18                  |
| RFI                                | 0.2486                 | <b>0.0056</b>         |
| Slope                              | 0.642                  | 0.269                 |



ecomorphological radiation was a global phenomenon, even as regional tectonic events such as the initial collision of the Indian subcontinent with Asia and Deccan Traps volcanism influenced local mammal evolution <sup>40,41</sup>.

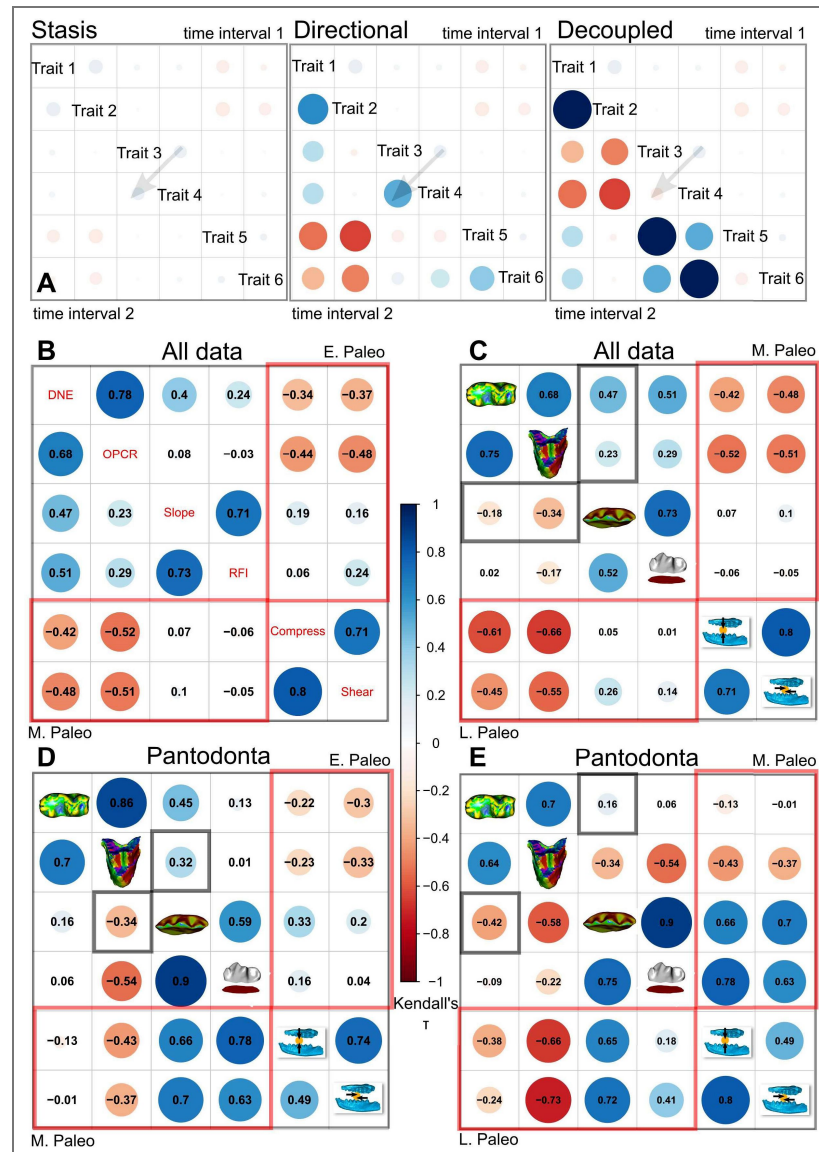
## Topography-performance integration underlies mammal dental shifts

Within an overall pattern of increasing integration between dental topographic traits and bite performance traits across the Paleocene time intervals (Fig. 3 <sup>42</sup>; Table S1 <sup>43</sup>), topographic versus performance trait variability shifted markedly both from early to middle and from middle to late Paleocene time bins (Figs S3-S4;). Early Paleocene to middle Paleocene DTA-FEA intra- and inter-correlations remained stable (Fig. 3B <sup>44</sup>), but DTA-FEA inter-correlations strengthened while intra-DTA correlations weakened in the Middle to Late Paleocene transition (Fig. 3C <sup>45</sup>). This transition pattern is in part due to the divergence of shape-performance linkages in CEP versus non-pantodont mammals. Dental topography variability tracked bite performance variability in non-pantodonts through time, peaking in the middle Paleocene (Fig S4D). By contrast, both intra- and inter-partition comparisons of topography and performance trends in CEPs showed a complete decoupling to shifts through time (Fig. S4G <sup>46</sup>). These findings indicate that form-function malleability, the coexistence of distinct topography-performance relationships in each time and taxon partition while overall integration between the two trait groups increases between time bins, was present throughout the Paleocene. Complex form-function linkages generally promote evolutionary redundancy and can enhance optimization of phenotypic traits when selective trade-offs are present <sup>42–44</sup>. The presence of functional redundancies underlies the high levels of dental topographic variability in Asian Paleocene mammals. Alternatively, varying degrees of independence between the two performance traits and dental topographic traits analyzed could allow the two aspects of the dentition to evolve in a decoupled manner (Fig. 3 <sup>47</sup>).

## South Asia as a Paleogene ‘Garden of Eden’ for placental mammals?

Among the most consequential implications of accurately interpreting post-K-Pg mammalian recovery dynamics in Asia is the ability to reconstruct the evolutionary conditions during the biogeographic origination of modern placental orders. Current knowledge and an overwhelming majority of data about the first post-dinosaur mammal ecological communities center on North American localities <sup>12,45</sup>, and secondarily from European records <sup>46</sup>. Yet, the fossil records of those continents pinpoint a high turnover of mammal faunas at the Paleocene-Eocene boundary, driven by the Paleocene origination of major modern mammal clades elsewhere and subsequent dispersal of those lineages to the European and North American continents, respectively. At least five living orders (Primates, Rodentia, Lagomorpha, Perissodactyla, Eulipotyphla), representing over half of mammalian species richness today, trace their evolutionary origins to Asia <sup>18,47–49</sup>. There is also mounting evidence that other organisms, including fish and plant lineages, followed a similar biogeographic pattern <sup>50</sup>.

The abrupt appearance of early representatives of modern mammal lineages in North America has been formally operationalized by the ‘East of Eden’ model, wherein Asia is the ‘Garden of Eden’ and a biodiversity pump of living orders of placental mammals <sup>51</sup> and other organisms. Biogeographic modeling analysis of modern mammalian diversity also strongly identifies tropical south Asia, including the geographic regions occupied today by southern China, India, and southeast Asian countries, as the cradle of mammalian diversification since the Paleocene <sup>52</sup>. As geographically proximate faunal assemblages to the epicenter of the ‘Garden of Eden’ hypothesis, the endemic Paleocene mammals of the Asian paleotropics analyzed in this study exhibited a high degree of dental topographic variability while strengthening topography-performance integration during a period of biogeographic isolation and regional and global climate changes. This finding favors the scenario whereby many living orders of placental mammals were borne out of phenotypically and functionally plastic ancestral assemblages, including those that lived in south China, during the Paleocene <sup>53</sup>.



**Fig. 3.** Correlation plots of dental topographic and bite performance traits in endemic Asian Paleocene mammals.

(A) Hypothetical correlation scenarios used to interpret stasis, directional, versus decoupled change through time in specimen data. (B) Pairwise ranked correlation coefficient estimated using Kendall's  $\tau$  between early and middle Paleocene dental topographic and performance traits in the main dataset. (C) Correlation between middle and late Paleocene traits in the main dataset. (D) Correlation between early and middle Paleocene traits in the Chinese endemic pantodont (CEP) data partition. (E) Correlation between middle and late Paleocene traits in the CEP data partition. Topography-performance correlations are marked in red boxes. Decoupled/reversed trait correlations are marked in gray boxes. Early-Middle

We further infer from these findings that episodes of fluctuating global warming during the first 10 Myr of the Cenozoic promoted the evolution of ‘all-purpose’ mammalian dentitions rather than those with specialized functions (Fig. 3 [↗](#)). This prolonged dental topographic variability tracks extended post-K-Pg floral recovery times [35,54](#), and suggests a ‘brawn before brains’ mode of placental mammal evolution may in part represent a series of correlated evolutionary shifts in sync with the steadily increasing complexity of Paleocene primary producers [20,36](#). That a global signal would be detected in the splendidly isolated south Asian Paleocene assemblage is unexpected, and indicates that effects of climate forcing on post-K-Pg mammal recovery may be ubiquitous, as has been predicted for Earth’s ongoing rapid environmental shifts [55](#). In response, Paleocene mammal clades in south China reached maximum tooth size disparity early and maximum integration between dental topography and bite performance later, all the while maintaining high levels of variability in dental complexity and convexity (Fig. 1 [↗](#)). These preconditions set the stage for the subsequent taxonomic turnover between archaic and crown lineages during the Paleogene to Neogene modernization of mammal communities [19,24](#). As a primary interface against shifting food resources and the environment, the placental dentition was poised to play an outsized role in the explosive diversification following the K-Pg extinctions, with the masticatory complex having already completed the macroevolutionary transition to a mechanically stiff but phenotypically flexible jaw during the Mesozoic Era [56,57](#). The end-Cretaceous extinctions and the climatic volatility that followed then set in motion the ecological release that allowed mammals to explore dental form-function to a far fuller extent in 66 Myr of the Cenozoic Era than during the preceding 150 Myr under the reign of dinosaurs. The transformational evolutionary sequence of ‘brawn before bite’ in tooth size, topographic, and performance evolution was likely central to this placental success story.

## Method details

We focused sampling on three clades: Chinese endemic Pantodonta, Chinese Arctostylopida, and Anagalida. Additional data were collected on other clades (e.g., Chinese Tillodontia) opportunistically when well-preserved specimens were available (Data S1 [↗](#)). These three main clades together represent >50% of the species found in Paleocene faunas across China [11](#). They also individually represent the most diverse clades across the three Asian Land Mammal Ages of the Paleocene: Shanghuan, Nongshanian, and Gashatan. By contrast, these clades are reduced to <25% of species diversity in Eocene assemblages in China, finally disappearing altogether in the late Oligocene. Therefore, we take the three clades to be representative of mammal assemblage dynamics during the Paleocene time interval in China.

Historically, Paleocene fossil mammal faunas in China have been defined based on biostratigraphic criteria and supplemented by magnetostratigraphic correlations where available. Here we follow the assessment of [59](#) in generally correlating the Shanghuan Asian Land Mammal Age (ALMA) with the Puercan and Torrejonian North American Land Mammal Ages (NALMAs), the Nongshanian ALMA with early to middle Tiffanian NALMAs, and the Gashatan ALMA with late Tiffanian and Clarkforkian NALMAs, respectively (see [11](#) for an alternative interpretation). Additionally, the Shanghuan and Nongshanian boundary was interpreted by [23](#) to be at the top of Chron C27N, which is dated at 60.920 Myr ago in the Geomagnetic Polarity Time Scale [60](#) but indicated as closer to ~62.3 Myr ago in [59](#). Similarly, the Nongshanian-Gashatan boundary has been variably defined at 59.24 Myr ago [59](#) or the base of Chron C26N, which is 57.911 Myr ago [60](#). In contrast, the end of the Gashatan coincides with the Paleocene-Eocene boundary at 56 Myr ago and has been consistently defined as such [59](#). Given these existing uncertainties, we use the terms ‘early Paleocene’, ‘middle Paleocene’, and ‘late Paleocene’ to refer generally to the Shanghuan, Nongshanian, and Gashatan ALMAs, respectively, in this study.

We analyzed 200 individual teeth from 48 specimens, representing 37 species (Data S1-S2 [↗](#)). The specimens represent 2 upper first premolars, 4 upper second premolars, 15 upper third premolars, 19 upper fourth premolars, 22 upper first molars, 23 upper second molars, 16 upper third molars, 3 lower first premolars, 5 lower second premolars, 13 lower third premolars, 17 lower fourth premolars, 20 lower first molars, 22 lower second molars, and 19 lower third molars.



The assigned geologic age of the studied specimens spans the entire Paleocene, binned into early (n = 91 teeth), middle (n = 46 teeth), and late Paleocene (n = 63 teeth) data partitions. To maximize sample size while minimizing disturbance to more delicate specimens, a combination of original specimens and previously produced high-fidelity specimen casts were used in our sampling of dental crown morphology.

## Tooth shape estimates using Dental Topographic Analysis

We digitized dental crown morphology via microCT either using a GE v|tome|x m 300/180 kV micro-computed-tomography system (GE Measurement & Control Solutions, Wunstorf, Germany), housed at the Institute of Vertebrate Paleontology and Paleoanthropology (IVPP), Chinese Academy of Sciences (CAS), or a GE Phoenix Nanotom M in the Functional Anatomy and Vertebrate Evolution Laboratory, University of California, Berkeley. Projection images (1,000 to 1,500 images depending on specimen size) were acquired with an isotropic voxel size of 10 to 40  $\mu\text{m}$ , at an energy range of 120–150 kV, current of 100–150  $\mu\text{A}$ . Additionally, surface 3D scans (generated using IVPP's Artec Space Spider 3D scanner, with a precision of 0.05 mm and resolution of 0.1 mm) were used for larger specimens to efficiently obtain surface morphology data. The enamel caps (the crown portion of each tooth above the enamel-dentine junction) were extracted from the specimen models using Geomagic Wrap v2021; all crown surface holes (generated from scanning imperfections or digitally removed sediment) were patched, the models were remeshed with triangles that have aspect ratios (base:height or edge-edge) of <10 and decimated to ~10,000 triangles. Finally, we standardized the spatial orientation of all tooth crowns before exporting them as .ply files for dental topographic analysis (DTA). Individual tooth crown (enamel cap) models have the Z axis normal to the occlusal plane, the X axis parallel to the mesial-distal axis, and the Y axis parallel to the labial-lingual axis.

Exported meshes were further vetted and prepared for DTA within the R programming environment. We used functions implemented in the *molarR* R package<sup>61</sup> for all of the steps described next. First, all .ply files are further cleaned using 'molarR\_Clean' to remove floating points/vertices and any triangular faces with zero area. Then, the batch function 'molarR\_Batch' was used to calculate four dental topographic metrics: DNE, OPCR, Slope, and RFI.

Dirichlet Normal Energy (DNE) is a measure of occlusal sharpness, using a quantification of surface energy in tooth crowns relative to a gently curving or flat mesh surface. Steep, high, and/or shearing cusps tend to produce higher DNE values in DTA, whereas bulbous cusps tend to produce lower DNE values<sup>62,63</sup>. DNE has been used to successfully distinguish folivores, omnivores, and frugivores in euarchontan (primates, colugos, tree shrews) mammals<sup>62,64</sup>.

Orientation Patch Count Rotated (OPCR) is a measure of tooth crown complexity, using quantification of distinct patches on the crown face that possess unique orientations. Teeth with larger number of cusps and crenulations tend to produce higher OPCR values, whereas teeth with fewer cusps and simpler ridges tend to produce lower OPCR values. This metric has been used effectively to distinguish convergently evolved carnivore versus herbivore species across multiple mammalian orders<sup>65</sup>. DeMers and Hunter<sup>66</sup> suggest that OPCR does not currently provide a reliable approximation for the degree of adaptation to herbivory, and that any interpretation of ecomorphological differences should be made within a taxon-specific context. Given the narrow set of clades targeted in our dataset, we interpret increasing OPCR as an indication of increased functional capability to process vegetation. A related measure, slope, quantifies the mean tilt of the tooth surface relative to the occlusal plane. Teeth with lower and more gently curving cusps tend to have lower slopes, whereas teeth with sharp and/or abruptly ridged crests tend to have higher slopes.

Relief Index (RFI) is a measure of the relative height and complexity of the tooth crown, using the values of the 3D surface and 2D 'footprint' areas of the tooth crown. RFI has been shown to be effective in distinguishing frugivores from insect and leaf specialists in euarchontan mammals<sup>62</sup>. Furthermore, RFI is also able to distinguish carnivoran species at different trophic levels and dietary breadths<sup>63</sup>.

All tooth crown models are provided as ply files and deposited in FigShare (10.6084/m9.figshare.28611854).

## Tooth performance estimates using Finite Element Analysis

The common inference that dental morphology can reflect dietary adaptations makes an underlying connection between the two traits through biomechanical performance; the mechanical performance imbued by a particular morphological configuration accomplishes one or more food acquisition and/or processing tasks. Such mechanical performance links can be tested using experimental and simulation approaches on both theoretical grounds and actual tooth shapes<sup>67</sup>. Here we applied a simulation approach to estimate two general performance traits commonly interpreted for mammalian tribosphenic teeth: puncturing/compressing and shearing/grinding<sup>68</sup>. The ancestral therian tribosphenic stroke, or the chewing movement underlying mammals with unfused mandibular symphysis and tribosphenic molars, has been reconstructed to involve significant components of (1) long-axis rotation in each hemimandible and (2) mortar-and-pestle grinding between upper and lower cusps<sup>69</sup>. These movements are associated with the ‘lock-and-key’ occlusion of a stereotypical mammalian tribosphenic dentition, which exhibit tall crest-like cusps in the trigonid and lower basin-like valleys in the talonid of lower molars that articulate with their corresponding upper teeth (Fig. S3<sup>70</sup>).

We used finite element analysis (FEA) to estimate the work efficiency of different tooth crowns subjected to compressive/puncturing and shearing/grinding forces, the two major masticatory actions of tribosphenic teeth<sup>68,70,71</sup>. Adequate compressive and puncturing forces applied through individual or combined action of cusps are needed to overcome the fracture strength of brittle and tough food such as seeds and invertebrate exoskeletons in order for individuals to access soft and/or more nutritious tissues within. Sufficient shearing and grinding forces applied through surfaces or cusp edges are necessary to further break down food boli into smaller pieces to aid in the digestive process. The combination of these two major functions of tribosphenic teeth is thought to be a foundational adaptation that enabled the mammalian radiation<sup>68,69,72</sup>. We focused on the dental enamel portion of the tooth crown in our biting simulations because it is the most mineralized vertebrate tissue and typically the best-preserved component of fossil teeth. The fracture mechanics of mammalian dental enamel is thought to be intimately related to dietary function<sup>73</sup>, including the evolution of different enamel thickness in different feeding ecomorphologies<sup>28</sup>. It is important to note that because the dental data available for this study included not only CT scans, but also surface scans and specimen casts, enamel thickness was not quantifiable for most specimens in our dataset (see below for a description of enamel thickness scaling procedure applied in the simulations).

The two masticatory scenarios tested, compressing/puncturing versus shearing/grinding, have previously been shown to be broadly connected to tooth topology. Taller cusps (approximated in our study by Relief Index and partially by Slope) and more convex crowns (approximated in our study by DNE) tend to exhibit higher strain in finite element models of hypothetical tooth shapes<sup>74</sup>. However, there is by no means a consensus on the presence of a tight form-function linkage across all tooth types and/or clades. Whereas a dental topography to food mechanical property linkage can be detected in some extant carnivores<sup>75,76</sup>, bats<sup>77</sup>, and primate-like tooth models<sup>78</sup>, a form-function relationship between dental topography and tooth/food mechanical properties is absent in other model systems<sup>64,79</sup>. Therefore, we incorporated biomechanical simulations as an explicit test of the mechanical performance associations implied by the DTA-flora correlations established in the dental topography and paleoenvironmental portion of our analyses. The main question addressed by these simulations is not necessarily whether a form-function linkage exists in the Paleocene mammal dataset, but whether there is a consistent relationship between DTA and FEA traits across the three time bins of the Paleocene. If such consistent linkages are observed, it would suggest a stasis in dental topography and compressive/shearing performance through the first 10 million years of the Cenozoic in our dataset; alternatively, any significant changes in the topography-performance relationship would indicate substantial evolutionary change through the same time period.

We imported the same tooth crown models generated for DTA into the Strand7 finite element software (Strand7 Pty Ltd, Australia) to create finite element models of the individual teeth. The tooth meshes are composed of two-dimensional, three-noded triangular elements ranging between 9,000-10,000 elements, with assigned thickness parameter (see below) so they behave similarly to single-layer 3-D elements. A convergence test was done on two of the specimens (IVPP V5228, V5231) and indicated that the <10,000 three-noded triangular plate elements (tri3) returned simulation output values with less than 10% deviation from higher resolution meshes of the same models (Fig. S3 [\[link\]](#)). Therefore, we deemed the default resolution of the tooth meshes used in DTA to be sufficient for finite element simulations. Furthermore, because the dataset contained digital models built from both CT and surface scan data, not all specimens have enamel thickness information. Although increased enamel thickness has been associated with durophagy in mammals, there appears to be taxon-specific patterns of enamel thickness to tooth size and dietary ecology, in addition to extensive variation in intra-tooth enamel thickness distributions [80–82](#). In absence of information on enamel thickness differences among the taxa studied, we standardized the thickness of the tooth crown models isometrically to be 10% of the total surface area of a given tooth model. As such, the simulation results should be treated as enamel cap shape-derived performance traits, rather than those based on a fully parameterized enamel model incorporating localized thickness and inter-specific allometric differences. We scaled applied forces on the teeth, simulating compressive or shearing loads, to be a value equivalent to total model surface area (but in units of Newtons instead of  $\text{mm}^2$ ). Compressive loads were represented on each tooth by a single nodal force vector directed towards the base of the crown along the height axis of each tooth, on the tallest cusp or cuspid (which is typically the protocone for upper molars and protoconid for lower molars, or the main cusp in premolars). This configuration simulated a compressive load directed into the tallest cusp (towards the root) on a given tooth from a food item (Fig S3), as reconstructed for the mammalian tribosphenic bite during a crushing or puncturing movement [68,69](#). Shearing loads were represented on each tooth by two nodal force vectors, one on each of the two tallest cusps or cuspids; the two nodal forces are of equal magnitude but opposing directions, perpendicular to the long axis of the tooth and in the occlusal plane. This configuration portrays a shearing motion along the short axis of the tooth (Fig. S3 [\[link\]](#)), and simulates tooth-to-tooth and tooth-to-food contact during the ‘mortar and pestle’ grinding rotation movement reconstructed for the ancestral mammal tribosphenic bite [69](#). This force magnitude standardization procedure effectively generates identical force to area ratios across the models, ensuring that the magnitudes of mechanical stress placed on the model are the same across models of different sizes; furthermore, we adjusted the strain energy values collected for each tooth model simulation using the volume and input force ratio-based correction equation provided in [83](#). Given our objective to assess the association between relative performance traits and DTA metrics, with the latter being size-free variables, only the relative magnitudes of bite performance, rather than absolute values, were collected from the performance analyses.

After each tooth crown model was defined using the criteria outlined above, homogenous material properties were assigned. All models were defined as plate models (with thickness scaled to surface area) with a Young’s (elastic) modulus of 80 GPa [84](#) and Poisson Ratio of 0.3. Next, each tooth was constrained from translation and rotation using four nodal constraints distributed in the four corners of each tooth, respectively (Fig. S3 [\[link\]](#)).

All bite scenarios were solved using the linear static solver implemented in Strand7. A total number of 400 analyses were performed (one compressive bite and one shearing bite simulation for each of the 200 tooth specimens in the dataset). We extracted two traits from the models: compressive bite strain energy and shear bite strain energy. Strain energy is defined as the area under a stress-strain curve of an object under load; operationally it measures the amount of work done by the load to deform the object (analogous to experimental work-to-fracture measurements)[85](#). An object that is more resistant to deformation would have lower strain energy than an object that easily deforms under load. The use of work-to-fracture measures to assess tooth performance is consistent with the understanding of mammalian tooth enamel as a fracture-prone, yet fracture-resistant, biological tissue [86](#). In such a framework, fracture resistance is expected be directly related to the amount of force an individual is able to exert through the tooth

crown during mastication, and by extension the hardest or toughest food item that can be processed without catastrophic damage to the tooth crown itself. Fracture resistance (as approximated by strain energy in this study) is also likely to be a stronger target of selection in mammals compared to those of other toothed vertebrates, given the former's diphyodonty (having only two sets of tooth instead of continuous dental replacement) and thus necessity to prolong usable tooth lifespan<sup>80</sup>. Therefore, we used strain energy values under compressive and shear bite simulation scenarios as a proxy for the effectiveness of each tooth crown model at resisting deformation from the respective biting forces. A tooth that is more resistant (i.e., have lower strain energy values) from compressive forces would be able to more effectively crush/puncture harder food items, and a tooth more resistant from shear forces would be able to cut/grind tougher food items. We modify Irschick et al.'s<sup>87</sup> definition of whole-organism performance and define bite performance in the context of this study as the capacity of individual tooth models to resist simulated compressive and shearing forces, which represent ecologically relevant factors that influence masticatory efficacy.

All finite element models are provided as NASTRAN/text files and deposited in FigShare (10.6084/m9.figshare.28611854).

## Sensitivity validation of original versus cast specimen models

The tooth dataset developed in this study contains a mixture of CT scans of original and casts of fossil specimens. In order to assess the potential discrepancies in DTA and FEA trait values collected from cast versus original specimen derived models, either due to phenotypic details not captured in cast specimens or deterioration of epoxy or resin-based casts over time, we randomly sampled two specimens for which both original and cast based models were analyzed. Results from DTA and FEA of the CT-derived models of lower first molars from IVPP V5228 (*Altibalambdactus*) and V5231 (*A. tenuis*) were compared to those conducted on digital models built from CT scans of casts of those specimens.

We used the 'n-point' and 'global' alignment functions in Geomagic Wrap to first align the models using arbitrary homologous landmarks visually identified on both tooth models and then the automatic global alignment function to maximize overlap between the two models in 3D coordinate space. The 'deviation' function in Geomagic Wrap was then used to calculate summary statistics for linear deviations orthogonal to the surface of the original model. The V5228 Lm1 model showed a maximum deviation of 0.58 mm (5.8% of crown height) and average deviation of 0.04-0.09 mm (<1% of crown height), with a standard deviation of 0.11 mm (1% of crown height). The V5231 Lm1 model showed a maximum deviation of 0.38 mm (6% of crown height) and average of 0.03-0.04 mm (<1% of crown height), with a standard deviation of 0.06 mm (1% of crown height)(Fig. S3 [S3](#)).

## Quantification of uncertainty ranges for bootstrap sampling

We then subjected the cast-based models to DTA and FEA. In both cases, the DTA values between cast and specimen models are the most different for DNE (30-40% difference), followed by OPCR (18-32% difference), then RFI (2-13% difference), and finally Slope (1-5% difference) (Data S4 [S4](#)). Also in both cases, adjusted compressive SE values differed by 14-41% and adjusted shear values differed by 20% (Data S3 [S3](#)). Based on these specimen-cast validation tests, we set a conservative +/-40% uncertainty range for all DTA and FEA values obtained from all cast-based tooth models. DTA and FEA trait mean and variance estimates used for downstream statistical analyses were then calculated using a bootstrap sampling scheme where 1,000 replicates of the DTA and FEA datasets were passed through the statistical tests (see Quantification and statistical analysis section, below). Overall, the early and middle Paleocene time bins contain around 50% cast data, whereas the late Paleocene time bin contains 66% cast data. If data uncertainty from original versus specimen models present a major signal, we would expect the late Paleocene time interval to always show higher variance given an abundance of cast data in that time bin. We did not

observe any consistent trends of high variance in the late Paleocene in the 1,000 bootstrap samples, suggesting that the results are not significantly biased by data quality differences between cast and original specimen derived models (Fig. S4 [↗](#)).

## Time bin duration correction and tooth position ratios

According to Ni et al. [6], the Shanghuan Asian Land Mammal Age (ALMA) faunas in Guangdong and Anhui range from 66–61.6 Ma (or 4.4 Myr in duration), the Nongshanian ALMA faunas in Guangdong, Jiangxi, and Anhui range from 61.6–59.2 Ma (or 2.4 Myr in duration), and the Gashatan ALMA faunas in Anhui ranges from 59.2 to 56 Ma (or 3.2 Myr in duration). Given the different durations represented in each of the three time bins used in our analyses, we corrected all variance estimates by dividing the variance calculated in each of the bootstrapped samples by the time duration of the respective time bin. In this manner, the variance values reported in the study represent per-million-year values.

We used pie chart analysis to verify that the proportions of tooth positions in each time bin are not substantially different from each other (Fig. S1 [↗](#)). However, we caution that time bin comparisons of individual tooth positions are unlikely to be statistically robust because of small sample sizes. We only use aggregates of dental positions (all teeth, molars, or premolars) in our data partition analyses and interpretation.

## Quantification and statistical analysis

We used the following features of the dental topography and performance data as the basis for assessing faunal assemblage trait shifts through the Paleocene: trait mean, variance, trait-to-trait correlation, and partition-to-partition correlation. We used ANOVA (Analysis of Variance) and pairwise t test to compare trait means, F test to compare trait variance, linear regression analysis and Kendall's  $\tau$  to compare trait-to-trait correlation, and two-block partial least squares (2B-PLS) analysis to compare partition-to-partition correlation. There are a total of six traits forming two main data partitions: topographic partition (DNE, OPCR, Slope, RFI) and performance partition (Compressive SE, and Shear SE). To assess the sensitivity of the results to subsets of the data, all of the trait comparisons were done iteratively using the total dataset, by time bin (early, middle, late Paleocene), taxon (Chinese endemic pantodonts versus non-pantodonts), and/or by dental position (all teeth, molar teeth, premolar teeth). All statistical tests were performed on 1,000 bootstrap resampled datasets that were parameterized using results of sensitivity tests on original versus cast specimen models, finite element mesh convergence tests, and time bin duration comparisons (see previous section). In addition, we quantified tooth size using two measures: total 2D surface area and square root of surface area. The R script for all statistical analyses and plots are included in [Data S9 \[↗\]\(#\)](#).

## Statistical resampling and tests

We used bootstrap resampled variance of dental topographic metrics and dental performance variables as a measure of tooth form-function disparity. 1,000 replicates of the 200-specimen DTA-FEA trait datasets were generated by sampling each DTA and FEA trait value from a uniform distribution defined using a  $\pm 40\%$  range of the original model-derived values. The bootstrap resampling was done with replacement, so it was possible for a given DTA/FEA trait value in a sampled tooth model to be repeated in a different replicate, but very unlikely for entire datasets to share similar values with another replicate. We used this resampling scheme to account for the uncertainty in trait estimates introduced by the totality of modeling, specimen preservation, cast versus original, and other potential sources of uncertainty in trait values. For each replicate, variance was calculated for each metric, time interval, for Chinese endemic pantodont (CEP) versus non-pantodont, and molar versus premolar data partitions. To assess shifts through time, statistical differences between the variance of each dental topographic metric sample in adjacent time intervals (A, early Paleocene; B, middle Paleocene; C, late Paleocene) were evaluated using F tests. A null hypothesis of a variance ratio of 1 between pairwise comparisons was tested using the `var.test()` function in R <sup>88</sup>. The outputs of the variance tests on a give bootstrap sample were then compiled for all 1,000 replicates, and the overall variance mean and variance test output



used as the statistical basis for detecting significant differences between time, taxon, and tooth data partitions. Unless indicated otherwise, all tests described below were also done using the 1,000 bootstrap samples.

We evaluated differences in central tendency (mean value) of dental topographic metrics and dental performance variables through time using analysis of variance (ANOVA) and pairwise *t* tests. Each dental topographic metric was tested against the three time intervals defined above, using the `aoV()` function in R. For statistically significant (at the  $p < 0.05$  level) results, we additionally evaluated the pairwise intervals that contribute significant differences in mean dental topographic values. We assessed pairwise differences using pairwise *t* tests implemented with a Holm correction for multiple comparisons. The *t* test was conducted using `pairwise.t.test()` in R.

We constructed a tooth morphospace using all four dental topographic metrics analyzed by principal components (PC) analysis. The first and second PC axes were chosen to visualize the two-dimensional morphospace. Additionally, we quantified the degree of morphological disparity and statistical differences in disparity between adjacent time interval data partitions. All PC scores generated from the PCA were included in the disparity analysis. The `prcomp()` function in R was used for PCA, and the `morphol.disparity()` function implemented in the Geomorph R package<sup>89</sup> was used for morphological disparity significance tests. Input data for the PCA were from the original dataset values, not from the bootstrap samples.

Linear regression analysis between individual dental topographic metric (DNE, OPCR, Slope, RFI) and dental performance metric (compressive bite strain energy, shear bite strain energy) was performed to quantify the correlation between dental form and function. Adjusted  $R^2$  and *p* values generated from the `lm()` function in R were used to evaluate the goodness of fit and statistical significance of the form-function relationships. The distribution of  $R^2$  values and their corresponding *p* values are reported in Fig. S6<sup>90</sup>.

In addition to pairwise form-function linear regression analyses, we also evaluated the degree of correlation between the DTA and FEA data blocks as a means to measure integration between dental topography and deformation resistance. We used two-block partial least squares analysis<sup>90</sup> coupled with bootstrap resampling to generate distributions of correlation coefficients (*r-pls*) for each of the three time bins, and then tested for significant differences in the magnitude of DTA-FEA correlation between adjacent time bins using Welch's two-sample *t* tests on the correlation coefficient distributions. DTA-FEA correlation differences at  $p < 0.05$  are interpreted as a significant shift in the degree of integration of the two traits from one time bin to the next time bin.

All R scripts used in the analyses are included as supplementary files (Data S8, S9<sup>91</sup>).

## Resource availability

- Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Jack Tseng ([zjt@berkeley.edu](mailto:zjt@berkeley.edu) <sup>92</sup>).
- Code associated with analyses is included with this article.
- Raw data are available in the supplementary materials.
- All original fossil specimens are accessioned in the Institute of Vertebrate Paleontology and Paleoanthropology, Chinese Academy of Sciences.

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Formal analyses: ZJT.

Investigation: All authors. Results acquisition: ZJT.

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## Additional files

**Document S1.** [Figures S1–S6, Tables S1 and S2, and supplemental references.](#)

**Data S1.** [Data S1.](#)

**Data S2.** [Data S2.](#)

**Data S3.** [Data S3.](#)

**Data S4.** [Data S4.](#)

**Data S5.** [Data S5.](#)

**Data S6.** [Data S6.](#)

**Data S7.** [Data S7.](#)

**Data S8.** [Data S8.](#)

**Data S9.** [Data S9.](#)

**Data S10.** [Data S10.](#)

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## Peer reviews

### Reviewer #1 (Public review):

#### Summary:

This work provides valuable new insights into the Paleocene Asian mammal recovery and diversification dynamics during the first ten million years post-dinosaur extinction. Studies that have examined the mammalian recovery and diversification post-dinosaur extinction have primarily focused on the North American mammal fossil record, and it's unclear if patterns documented in North America are characteristic of global patterns. This study examines dietary metrics of Paleocene Asian mammals and found that there is a body size disparity increase before dietary niche expansion and that dietary metrics track climatic and paleobotanical trends of Asia during the first 10 million years after the dinosaur extinction.

#### Strengths:

The Asian Paleocene mammal fossil record is greatly understudied, and this work begins to fill important gaps. In particular, the use of interdisciplinary data (i.e., climatic and paleobotanical) is really interesting in conjunction with observed dietary metric trends.

#### Weaknesses:

While this work has the potential to be exciting and contribute greatly to our understanding of mammalian evolution during the first 10 million years post-dinosaur extinction, the major weakness is in the dental topographic analysis (DTA) dataset.

There are several specimens in Figure 1 that have broken cusps, deep wear facets, and general abrasion. Thus, any values generated from DTA are not accurate and cannot be used to support their claims. Furthermore, the authors analyze all tooth positions at once, which makes this study seem comprehensive (200 individual teeth), but it's unclear what sort of noise this introduces to the study. Typically, DTA studies will analyze a singular tooth position (e.g., Pampush et al. 2018 Biol. J. Linn. Soc.), allowing for more meaningful comparisons and an understanding of what value differences mean. Even so, the dataset consists of only 48 specimens. This means that even if all the specimens were pristinely preserved and generated DTA values could be trusted, it's still only 48 specimens (representing 4 different clades) to capture patterns across 10 million years. For example, the authors note that their

results show an increase in OPCR and DNE values from the middle to the late Paleocene in pantodonts. However, if a singular tooth position is analyzed, such as the lower second molar, the middle and late Paleocene partitions are only represented by a singular specimen each. With a sample size this small, it's unlikely that the authors are capturing real trends, which makes the claims of this study highly questionable.

<https://doi.org/10.7554/eLife.108917.1.sa2>

## Reviewer #2 (Public review):

### Summary:

This study uses dental traits of a large sample of Chinese mammals to track evolutionary patterns through the Paleocene. It presents and argues for a 'brawn before bite' hypothesis - mammals increased in body size disparity before evolving more specialized or adapted dentitions. The study makes use of an impressive array of analyses, including dental topographic, finite element, and integration analyses, which help to provide a unique insight into mammalian evolutionary patterns.

### Strengths:

This paper helps to fill in a major gap in our knowledge of Paleocene mammal patterns in Asia, which is especially important because of the diversification of placentals at that time. The total sample of teeth is impressive and required considerable effort for scanning and analyzing. And there is a wealth of results for DTA, FEA, and integration analyses. Further, some of the results are especially interesting, such as the novel 'brawn before bite' hypothesis and the possible link between shifts in dental traits and arid environments in the Late Paleocene. Overall, I enjoyed reading the paper, and I think the results will be of interest to a broad audience.

### Weaknesses:

I have four major concerns with the study, especially related to the sampling of teeth and taxa, that I discuss in more detail below. Due to these issues, I believe that the study is incomplete in its support of the 'brawn before bite' hypothesis. Although my concerns are significant, many of them can be addressed with some simple updates/revisions to analyses or text, and I try to provide constructive advice throughout my review.

(1) If I understand correctly, teeth of different tooth positions (e.g., premolars and molars), and those from the same specimen, are lumped into the same analyses. And unless I missed it, no justification is given for these methodological choices (besides testing for differences in proportions of tooth positions per time bin; L902). I think this creates some major statistical concerns. For example, DTA values for premolars and molars aren't directly comparable (I don't think?) because they have different functions (e.g., greater grinding function for molars). My recommendation is to perform different disparity-through-time analyses for each tooth position, assuming the sample sizes are big enough per time bin. Or, if the authors maintain their current methods/results, they should provide justification in the main text for that choice.

Also, I think lumping teeth from the same specimen into your analyses creates a major statistical concern because the observations aren't independent. In other words, the teeth of the same individual should have relatively similar DTA values, which can greatly bias your results. This is essentially the same issue as phylogenetic non-independence, but taken to a much greater extreme.

It seems like it'd be much more appropriate to perform specimen-level analyses (e.g., Wilson 2013) or species-level analyses (e.g., Grossnickle & Newham 2016) and report those results in

the main text. If the authors believe that their methods are justified, then they should explain this in the text.

(2) Maybe I misunderstood, but it sounds like the sampling is almost exclusively clades that are primarily herbivorous/omnivorous (Pantodonta, Arctostylopida, Anagalida, and maybe Tillodontia), which means that the full ecomorphological diversity of the time bins is not being sampled (e.g., insectivores aren't fully sampled). Similarly, the authors say that they "focused sampling" on those major clades and "Additional data were collected on other clades ... opportunistically" (L628). If they favored sampling of specific clades, then doesn't that also bias their results?

If the study is primarily focused on a few herbivorous clades, then the Introduction should be reframed to reflect this. You could explain that you're specifically tracking herbivore patterns after the K-Pg.

(3) There are a lot of topics lacking background information, which makes the paper challenging to read for non-experts. Maybe the authors are hindered by a short word limit. But if they can expand their main text, then I strongly recommend the following:

(a) The authors should discuss diets. Much of the data are diet correlates (DTA values), but diets are almost never mentioned, except in the Methods. For example, the authors say: "An overall shift towards increased dental topographic trait magnitudes ..." (L137). Does that mean there was a shift toward increased herbivory? If so, why not mention the dietary shift? And if most of the sampled taxa are herbivores (see above comment), then shouldn't herbivory be a focal point of the paper?

(b) The authors should expand on "we used dentitions as ecological indicators" (L75). For non-experts, how/why are dentitions linked to ecology? And, again, why not mention diet? A strong link between tooth shape and diet is a critical assumption here (and one I'm sure that all mammalogists agree with), but the authors don't provide justification (at least in the Introduction) for that assumption. Many relevant papers cited later in the Methods could be cited in the Introduction (e.g., Evans et al. 2007).

(c) Include a better introduction of the sample, such as explicitly stating that your sample only includes placentals (assuming that's the case) and is focused on three major clades. Are non-placentals like multituberculates or stem placentals/eutherians found at Chinese Paleocene fossil localities and not sampled in the study, or are they absent in the sampled area?

(d) The way in which "integration" is being used should be defined. That is a loaded term which has been defined in different ways. I also recommend providing more explanation on the integration analyses and what the results mean.

If the authors don't have space to expand the main text, then they should at least expand on the topics in the supplement, with appropriate citations to the supplement in the main text.

(4) Finally, I'm not convinced that the results fully support the 'brawn before bite' hypothesis. I like the hypothesis. However, the 'brawn before ...' part of the hypothesis assumes that body size disparity (L63) increased first, and I don't think that pattern is ever shown. First, body size disparity is never reported or plotted (at least that I could find) - the authors just show the violin plots of the body sizes (Figures 1B, S6A). Second, the authors don't show evidence of an actual increase in body size disparity. Instead, they seem to assume that there was a rapid diversification in the earliest Paleocene, and thus the early Paleocene bin has already "reached maximum saturation" (L148). But what if the body size disparity in the latest Cretaceous was the same as that in the Paleocene? (Although that's unlikely, note that papers like Clauser & Redner 2009 and Grossnickle & Newham 2016 found evidence of greater body size disparity in the latest Cretaceous than is commonly recognized.) Similarly, what if body

size disparity increased rapidly in the Eocene? Wouldn't that suggest a 'BITE before brawn' hypothesis? So, without showing when an increase in body size diversity occurred, I don't think that the authors can make a strong argument for 'brawn before [insert any trait]'.

Although it's probably well beyond the scope of the study to add Cretaceous or Eocene data, the authors could at least review literature on body size patterns during those times to provide greater evidence for an earliest Paleocene increase in size disparity.

<https://doi.org/10.7554/eLife.108917.1.sa1>

## Author response:

### eLife Assessment

*This important study fills a major geographic and temporal gap in understanding Paleocene mammal evolution in Asia and proposes an intriguing "brawn before bite" hypothesis grounded in diverse analytical approaches. However, the findings are incomplete because limitations in sampling design - such as the use of worn or damaged teeth, the pooling of different tooth positions, and the lack of independence among teeth from the same individuals - introduce uncertainties that weaken support for the reported disparity patterns. The taxonomic focus on predominantly herbivorous clades also narrows the ecological scope of the results. Clarifying methodological choices, expanding the ecological context, and tempering evolutionary interpretations would substantially strengthen the study.*

We thank Dr. Rasmann for the constructive evaluation of our manuscript. Considering the reviewers' comments, we plan to implement revisions to our study focusing on (1) expansion of the fossil sample description, including a detailed account of the process of excluding extremely worn or damaged teeth from all analyses, (2) expanded reporting of the analyses done on individual tooth positions, and tempering the interpretation of the pooled samples in light of the issues raised by reviewers, (3) providing a more comprehensive introduction that includes an overview of the Paleocene mammal faunas in south China, which unevenly samples certain clades whereas others are extremely rare, and why the current available fossil samples would not permit a whole-fauna analysis to be adequately conducted across the three land mammal age time bins of the Paleocene in China. We believe these revisions would substantially strengthen the study's robustness and impact for understanding the ecomorphological evolution of the earliest abundant placental mammals during the Paleocene in Asia.

### Public Reviews:

#### Reviewer #1 (Public review):

##### Summary:

*This work provides valuable new insights into the Paleocene Asian mammal recovery and diversification dynamics during the first ten million years post-dinosaur extinction. Studies that have examined the mammalian recovery and diversification post-dinosaur extinction have primarily focused on the North American mammal fossil record, and it's unclear if patterns documented in North America are characteristic of global patterns. This study examines dietary metrics of Paleocene Asian mammals and found that there is a body size disparity increase before dietary niche expansion and that dietary metrics track climatic and paleobotanical trends of Asia during the first 10 million years after the dinosaur extinction.*

##### Strengths:



*The Asian Paleocene mammal fossil record is greatly understudied, and this work begins to fill important gaps. In particular, the use of interdisciplinary data (i.e., climatic and paleobotanical) is really interesting in conjunction with observed dietary metric trends.*

**Weaknesses:**

*While this work has the potential to be exciting and contribute greatly to our understanding of mammalian evolution during the first 10 million years post-dinosaur extinction, the major weakness is in the dental topographic analysis (DTA) dataset.*

*There are several specimens in Figure 1 that have broken cusps, deep wear facets, and general abrasion. Thus, any values generated from DTA are not accurate and cannot be used to support their claims. Furthermore, the authors analyze all tooth positions at once, which makes this study seem comprehensive (200 individual teeth), but it's unclear what sort of noise this introduces to the study. Typically, DTA studies will analyze a singular tooth position (e.g., Pampush et al. 2018 Biol. J. Linn. Soc.), allowing for more meaningful comparisons and an understanding of what value differences mean. Even so, the dataset consists of only 48 specimens. This means that even if all the specimens were pristinely preserved and generated DTA values could be trusted, it's still only 48 specimens (representing 4 different clades) to capture patterns across 10 million years. For example, the authors note that their results show an increase in OPCR and DNE values from the middle to the late Paleocene in pantodonts. However, if a singular tooth position is analyzed, such as the lower second molar, the middle and late Paleocene partitions are only represented by a singular specimen each. With a sample size this small, it's unlikely that the authors are capturing real trends, which makes the claims of this study highly questionable.*

We thank Reviewer 1 for their careful review of our manuscript. A major external limitation of the application of DTA to fossil samples is the availability of specimens. Whereas a typical study design using extant or geologically younger/more abundant fossil species would preferably sample much larger quantities of teeth from each treatment group (time bins, in our case), the rarity of well-preserved Paleocene mammalian dentitions in Asia necessitates the analysis of small samples in order to make observations regarding major trends in a region and time period otherwise impossible to study (see Chow et al. 1977). That said, we plan to clarify methodological details in response to the reviewer's comments, including a more comprehensive explanation of our criteria for exclusion of broken tooth crowns from the analyses. We also plan to expand our results reporting on individual tooth position analysis, potentially including resampling and/or simulation analyses to assess the effect of small and uneven samples on our interpretation of results. Lastly, we plan to revise the discussion and conclusion accordingly, including more explicit distinction between well-supported findings that emerge from various planned sensitivity analyses, versus those that are more speculative and tentative in nature.

Chow, M., Zhang, Y., Wang, B., and Ding, S. (1977). Paleocene mammalian fauna from the Nanxiong Basin, Guangdong Province. *Paleontol. Sin. New Ser. C* 20, 1–100.

**Reviewer #2 (Public review):**

**Summary:**

*This study uses dental traits of a large sample of Chinese mammals to track evolutionary patterns through the Paleocene. It presents and argues for a 'brawn before bite' hypothesis - mammals increased in body size disparity before evolving more specialized or adapted dentitions. The study makes use of an impressive array of analyses, including dental topographic, finite element, and integration analyses, which help to provide a unique insight into mammalian evolutionary patterns.*

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We thank Reviewer 2 for raising several issues worthy of clarification. Separate analyses for individual tooth positions were performed but not emphasized in the first version of the study. In our revised version we plan to highlight the nuances of the results from premolar versus molar partition analyses.

*Also, I think lumping teeth from the same specimen into your analyses creates a major statistical concern because the observations aren't independent. In other words, the teeth of the same individual should have relatively similar DTA values, which can greatly bias your results. This is essentially the same issue as phylogenetic non-independence, but taken to a much greater extreme.*

*It seems like it'd be much more appropriate to perform specimen-level analyses (e.g., Wilson 2013) or species-level analyses (e.g., Grossnickle & Newham 2016) and report those results in the main text. If the authors believe that their methods are justified, then they should explain this in the text.*

We plan to emphasize individual tooth position analyses in our revisions, and provide a stronger justification for our current treatment of multiple teeth from the same individual specimens as independent samples. We recognize the statistical nonindependence raised by Reviewer 2, but we would point out that from an ecomorphological perspective, it is unclear to us that the heterodont dentition of these early Cenozoic placental mammals should represent a single ecological signal (and thus warrant using only a single tooth position as representative of an individual's DTA values). We plan to closely examine the nature of nonindependence in the DTA data within individuals, to assess a balanced approach to maximize information content from the relatively small and rare fossil samples used, while minimizing signal nonindependence across the dentition.

(2) Maybe I misunderstood, but it sounds like the sampling is almost exclusively clades that are primarily herbivorous/omnivorous (Pantodonta, Arctostylopida, Anagalida, and maybe Tillodonta), which means that the full ecomorphological diversity of the time bins is not being sampled (e.g., insectivores aren't fully sampled). Similarly, the authors say that they "focused sampling" on those major clades and "Additional data were collected on other clades ... opportunistically" (L628). If they favored sampling of specific clades, then doesn't that also bias their results?

If the study is primarily focused on a few herbivorous clades, then the Introduction should be reframed to reflect this. You could explain that you're specifically tracking herbivore patterns after the K-Pg.

We plan to revise the introduction section to more accurately reflect the emphasis on those clades. However, we would note that conventional dietary ecomorphology categories used to characterize later branching placental mammals are likely to be less informative when applied to their Paleocene counterparts. Although there are dental morphological traits that began to characterize major placental clades during the Paleocene, distinctive dietary ecologies have not been demonstrated for most of the clade representatives studied. Thus, insectivory was probably not restricted to "Insectivora", nor carnivory to early Carnivormorpha or "Creodonta", each of which represented less than 5% of the taxonomic richness during the Paleocene in China (Wang et al. 2007).

Wang, Y., Meng, J., Ni, X., and Li, C. (2007). Major events of Paleogene mammal radiation in China. *Geol. J.* 42, 415–430.

(3) There are a lot of topics lacking background information, which makes the paper challenging to read for non-experts. Maybe the authors are hindered by a short word limit. But if they can expand their main text, then I strongly recommend the following:

(a) The authors should discuss diets. Much of the data are diet correlates (DTA values), but diets are almost never mentioned, except in the Methods. For example, the authors say: "An overall shift towards increased dental topographic trait magnitudes ..." (L137). Does that mean there was a shift toward increased herbivory? If so, why not mention the dietary shift? And if most of the sampled taxa are herbivores (see above comment), then shouldn't herbivory be a focal point of the paper?

We plan to revise the text to make clearer connections between DTA and dietary inferences, and at the same time advise caution in making one-to-one linkages between them. Broadly speaking, dental indices such as DTA are phenotypic traits, and as in other phenotypic traits, the strength of structure-function relationships needs to be explicitly established before dietary ecological inferences can be confidently made. There is, to date, no consistent connection between dental topology and tooth use proxies and biomechanical traits in extant non-herbivorous species (e.g., DeSantis et al. 2017, Tseng and DeSantis 2024), and in our analyses, FEA and DTA generally did not show strong correlations to each other. Thus, we plan to continue to exercise care in interpreting DTA data as dietary data.

DeSantis LRG, Tseng ZJ, Liu J, Hurst A, Schubert BW, Jiangzuo Q. Assessing niche conservatism using a multiproxy approach: dietary ecology of extinct and extant spotted hyenas. *Paleobiology*. 2017;43(2):286-303. doi:10.1017/pab.2016.45

Tseng ZJ, DeSantis LR. Relationship between tooth macrowear and jaw morphofunctional traits in representative hypercarnivores. *PeerJ*. 2024 Nov 11;12:e18435.

(b) The authors should expand on "we used dentitions as ecological indicators" (L75). For non-experts, how/why are dentitions linked to ecology? And, again, why not mention diet? A strong link between tooth shape and diet is a critical assumption here (and one

*I'm sure that all mammalogists agree with), but the authors don't provide justification (at least in the Introduction) for that assumption. Many relevant papers cited later in the Methods could be cited in the Introduction (e.g., Evans et al. 2007).*

Thank you for this suggestion. We plan to expand the introduction section to better contextualize the methodological basis for the work presented.

*(c) Include a better introduction of the sample, such as explicitly stating that your sample only includes placentals (assuming that's the case) and is focused on three major clades. Are non-placentals like multituberculates or stem placentals/eutherians found at Chinese Paleocene fossil localities and not sampled in the study, or are they absent in the sampled area?*

We thank Reviewer 2 for raising this important point worthy of clarification. Multituberculates are completely absent from the first two land mammal ages in the Paleocene of Asia, and non-placentals are rare in general (Wang et al. 2007). We plan to provide more context for the taxonomic sampling choices made in the study.

Wang, Y., Meng, J., Ni, X., and Li, C. (2007). Major events of Paleogene mammal radiation in China. *Geol. J.* 42, 415–430.

*(d) The way in which "integration" is being used should be defined. That is a loaded term which has been defined in different ways. I also recommend providing more explanation on the integration analyses and what the results mean.*

*If the authors don't have space to expand the main text, then they should at least expand on the topics in the supplement, with appropriate citations to the supplement in the main text.*

We plan to clarify our usage of “integration” to enable readers to accurately interpret what we mean by it.

*(4) Finally, I'm not convinced that the results fully support the 'brawn before bite' hypothesis. I like the hypothesis. However, the 'brawn before ...' part of the hypothesis assumes that body size disparity (L63) increased first, and I don't think that pattern is ever shown. First, body size disparity is never reported or plotted (at least that I could find) - the authors just show the violin plots of the body sizes (Figures 1B, S6A). Second, the authors don't show evidence of an actual increase in body size disparity. Instead, they seem to assume that there was a rapid diversification in the earliest Paleocene, and thus the early Paleocene bin has already "reached maximum saturation" (L148). But what if the body size disparity in the latest Cretaceous was the same as that in the Paleocene? (Although that's unlikely, note that papers like Clauset & Redner 2009 and Grossnickle & Newham 2016 found evidence of greater body size disparity in the latest Cretaceous than is commonly recognized.) Similarly, what if body size disparity increased rapidly in the Eocene? Wouldn't that suggest a 'BITE before brawn' hypothesis? So, without showing when an increase in body size diversity occurred, I don't think that the authors can make a strong argument for 'brawn before [insert any trait]'.*

*Although it's probably well beyond the scope of the study to add Cretaceous or Eocene data, the authors could at least review literature on body size patterns during those times to provide greater evidence for an earliest Paleocene increase in size disparity.*

We plan to provide a broader discussion and any supporting evidence from the Cretaceous and Eocene to either make a stronger case for “brawn before bite”, or to refine what we mean by brawn/size/size disparity.

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